

Program: Diploma in Artificial Intelligence	
Course Code: 6349C	Course Title: Computer Vision and Image Processing Lab
Semester: 6	Credits: 1.5
Course Category: Program Elective	
Periods per week: 3 (L:0 T:0 P:3)	Periods per semester: 45

Course Objectives:

- To understand and implement image processing techniques.
- To understand image acquisition and storage using open-source software.
- To study, analyze and apply different techniques and algorithms for image processing.

Course Prerequisites:

Topic	Course code	Course name	Semester
Computer Programming concepts		Python Programming Lab	III
		Artificial Intelligence Lab	III
		Machine learning Lab	IV

Course Outcomes:

On completion of the course, the student will be able to:

CO n	Description	Duration (Hours)	Cognitive Level
CO1	Develop programs to implement image transformation in MATLAB	8	Applying
CO2	Develop programs to implement image enhancement.	9	Applying
CO3	Apply different Image Smoothing & Sharpening algorithms.	12	Applying
CO4	Apply different algorithms for image restoration and compression techniques	13	Applying
	Lab Exam	3	

CO – PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3						
CO2	2						
CO3	3						
CO4	3	3	3	3	3	3	3

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Develop programs to implement image transformation in MATLAB		
M1.01	Develop a program to display colour Images, find their complement and convert them to grey scale	2	Applying
M1.02	Implement Relationships between Pixels (To find Neighbour of a given Pixel)	2	Applying
M1.03	Implement transformations of an Image (Scaling, Rotation)	2	Applying
CO2	Develop programs to implement image enhancement.		
M2.01	Develop a program to perform contrast stretching of a low contrast image, Histogram	3	Applying
M2.02	Apply histogram equalization for enhancing the given images.	2	
M2.03	Analyze the rotation and convolution properties of the Fourier transform using any gray scale image.	2	Applying
M2.04	Display of bit planes of an Image.	2	Applying
	Lab Exam – I	1.5	
CO3	Apply different Image Smoothing & Sharpening algorithms		

M3.01	Develop a program to smoothen the given image using different spatial filters and study the effects.	3	Applying
M3.02	Develop programs to sharpen the given image using different spatial filters and study the effects.	3	Applying
M3.03	Develop programs to perform noise removal using different spatial filters and compare their performances.	3	Applying
M3.04	Develop programs to perform edge detection using different operators and compare the results.	3	Applying
CO4	Apply different algorithms for image restoration and compression techniques		
M4.01	Develop programs to study the effect of inverse filtering in restoring degraded images.	3	Applying
M4.02	Develop programs to perform compression techniques of the given image.	4	Applying
M4.03	**Open Ended Experiments	6	Applying
	Lab Exam -2	1.5	

**** For example –**

1. Real-Time Face detector .
2. Real-Time Color Detection.

Text / Reference

T/R	Book Title/Author
T1	“A Guide to MATLAB, 3e: For Beginners and Experienced Users” Brian R. Hunt, Ronald L. Lipsman, Jonathan M. Rosenberg
R1	MATLAB for Engineers, 6th edition Holly Moore, Salt Lake Community College Pearson Education Inc, 2022

Online Resources

Sl.No	Website Link
1	https://www.vtupulse.com/
2	https://in.mathworks.com